From Weyl Anomaly to Universal Surface Defect Entropy and Casimir Energy

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Brief Summary

Weyl anomaly (central charge) governs the entanglement structure and ground state energy of CFTs

Questions

- Is this relation still valid in the presence of defects?
- Any non-perturbative way to study defects?

Main results

- **Closed formulae** for surface defect contribution to twisted Rényi entropy and Casimir energy, **controlled by defect Weyl anomaly**
- **Non-perturbative methods** combining holography, localization and anomaly polynomial

Surface Defect Weyl Anomaly

Coefficients of the trace of the defect localized stress tensor

$$\langle t^\mu_\mu \rangle = -\frac{1}{24\pi}\left[\, b\, R^\Sigma + d_1 \Pi^\mu_{ab}\Pi^{ab}_\mu - d_2 W^{ab}_{ab}\right]$$

Defects in Higher-dimensional CFT

We study surface defect in 6d (2,0) SCFT

- Partition function of 6d (2,0) SCFT in radial quantization

$$Z = \mathrm{Tr}\left[e^{-\beta\left(\sum_{i=1}^3 \omega_i J_i + \sum_{a=1}^2 \sigma_a R_a\right)}\right]$$

- ½-BPS surface defect is labelled by its representation $\mathcal{R}$ with highest weight λ of type-A Lie algebra $\mathfrak{su}(N)$

$$b = 24(\lambda,\rho) + 3(\lambda,\lambda) \qquad d_2 = 24(\lambda,\rho) + 6(\lambda,\lambda)$$

- **Defect contribution to the SUSY Rényi entropy (dSRE)**

$$S_n = \frac{1}{1-n}\log\frac{Z_n}{(Z_1)^n} \qquad n\text{: Rényi index}$$

- **Defect contribution to Casimir energy (dSCE)** can be extracted from the low-temperature asymptotics of defect expectation value

$$\log\langle D\rangle = -\beta E_{\mathcal{R}} + O(\beta^0), \qquad \beta \to \infty$$

Comprehensive & Non-perturbative Methods

Holography

The large N behavior of dSRE

Surface defect in fundamental representation

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Probe M2-brane in 7d topological black hole

$$\log\langle D\rangle = -I_{M2} = \pi n T_2 r_H^2\, \mathrm{Vol}[\mathbb{H}^1]$$

$$S_n(N\to\infty) = 2N\left[\frac{r_1 r_2}{2}\left(\frac{1}{n}-1\right)+1\right]\log\frac{l}{\epsilon}$$

Anomaly Polynomial

Exact results of dSCE and dSRE

Consistency check

$$A_4 = \frac{1}{2}(\lambda,\lambda)\big(c_2(L)-c_2(R)\big) + (\lambda,\rho)\big(c_2(I)-c_2(F)\big)$$

Equivariant integral $\quad E_{\mathcal{R}} = -\int_{\mathbb{R}^2} A_4$

Extended integral $\quad \tilde{E} = -\int_{\mathbb{R}^2\times\mathbb{R}^2} A_4$

$$S_n = \frac{n\tilde{E}|_{n=1} - \tilde{E}|_n}{1-n}\mathrm{Vol}[\mathbb{H}^1]$$

Localization

The large N behavior of dSCE & $n\to 0$ behavior of dSRE

Dimensional reduction $\Longrightarrow$ 5d matrix integral with Wilson loop

$$\langle D\rangle = \frac{1}{Z}\int \mathrm{d}v\, \exp\left[\frac{2\pi}{\omega_1\omega_2\omega_3}\left(-\frac{\pi}{\beta}\sum_{i=1}^N v_i^2 + \frac{\sigma_1\sigma_2}{2}\sum_{j<i}(v_i - v_j)\right)\right]\mathrm{Tr}_{\mathcal{R}}\exp\left[\frac{2\pi v}{\omega_1}\right]$$

Saddle point approximation

$$E_{(k)} = -\frac{1}{2\omega_1}\left[k^2\,\omega_2\omega_3 + k(N-1)\,\sigma_1\sigma_2\right]$$

$$E_{[k]} = -\frac{1}{2\omega_1}\left[k\,\omega_2\omega_3 + k(N-k)\,\sigma_1\sigma_2\right]$$

dSCE governs dSRE$|_{n\to 0}$

$$S_{n\to 0} = -\frac{\mathrm{Vol}[\mathbb{H}^1]}{\mathrm{Vol}[S^1_\beta]}\beta E_{\mathcal{R}}\Big|_{n\to 0} = \frac{2b-d_2}{12}\frac{r_1 r_2}{n}\log\frac{l}{\epsilon}$$

Closed formulae

$$S_n = \frac{2b-d_2}{6}\left[\frac{r_1 r_2}{2}\left(\frac{1}{n}-1\right)+1\right]\log\frac{l}{\epsilon}$$

$$E_{\mathcal{R}} = -\frac{1}{\omega_1}\left[\frac{d_2-b}{6}\omega_2\omega_3 + \frac{2b-d_2}{24}\sigma_1\sigma_2\right]$$

Reference

Z.-X. Huang, M.-K. Yuan and Y. Zhou, From Weyl Anomaly to Defect Supersymmetric Rényi Entropy and Casimir Energy, [arXiv: 2501.09498].

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